

# Emma N. Bennett

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## Education

PhD in Computer Science

Anticipated Graduation Date: May 2028

MS in Computer Science, Data Science Certificate GPA: 4.0

University of Missouri, Columbia, Missouri, May 2025

BS in Chemical Engineering, Minor in Math GPA: 3.83

University of Missouri, Columbia, Missouri, May 2019

## Job Experience

Research Assistant, University of Missouri, Columbia, Missouri

January 2024 – Present

- Utilization of neural network architectures (CNN, ViT, GCN, etc.) to solve computer vision problems
- Implementation of graph analysis algorithms to perform approximate subgraph matching tasks
- Analysis of soybean crop canopy from point cloud data

Research Intern, MU Quantum Innovation Center, Columbia, Missouri

June 2025 – August 2025

- Developed an end-to-end hybrid quantum-classical graph neural network pipeline for aerial localization
- Implemented and benchmarked quantum projection layers on real IBM hardware and simulated backends
- Designed a large-scale sliding-window retrieval and perturbation testbed (~8.6K candidate windows), demonstrating 10-20 point Recall@1/5 gains over classical baselines under moderate-heavy graph noise

Teaching Assistant, University of Missouri, Columbia, Missouri

August 2024 – Present

- Assist students with understanding key course concepts during office hours
- Communicated with TA's and professors to maintain clear and consistent expectations for students
- Covered material such as linked lists, jagged arrays, graph search, recursion, dynamic programming

Staff Chemical Engineer, Burns & McDonnell, Kansas City, Missouri

December 2022 – April 2023

- Bid and purchase of mechanical equipment packages for drinking water and sewer costing upwards of \$1M
- Development of potable drinking water and sanitary sewer distribution systems across a 1,000+ acre site
- Collaboration with site civil to design gravity storm water collection system, segregating storm water based on environmental requirements for treatment, design of pellet collection system for storm water using Stokes' Law

Assistant Chemical Engineer, Burns & McDonnell, Kansas City, Missouri

June 2019 – December 2022

- Design wastewater treatment for high strength, high oxygen demand water through use of technologies such as primary oil/water separation, secondary oil/water separation, anaerobic, and aerobic treatment
- Use of hydraulic institute principles and Fathom to design below grade lift stations and pumping systems
- Mentor to summer interns: coordinating regular workload check-ins, delegating tasks, and providing support

Environmental Engineering Intern, Department of Natural Resources, Jefferson City, Missouri

May 2017– August 2017

- Updated the Wastewater Construction Permitting Manual and CAFCom cost tracking spreadsheet
- Reviewed sewer extensions using fluid and pipe calculations gaining knowledge of the NPDES permitting

Undergraduate Researcher, University of Missouri, Columbia, Missouri

September 2015— May 2017

- Conducted research in the field of biomedical optics use
- Analyzed laser output by applying LabVIEW programming skills
- Applied skills in 3D printing and modeling software to print pieces integral to optics projects

## Skills and Certifications

- Languages: C, Python, Java, R, MatLab, PyTorch, HTML, CSS.
- Libraries / Software: Python (Numpy, SciKit-Learn, OpenCV, PyTorch, Matplotlib, NetworkX, SciPy, Pandas, Qiskit, PennyLane), R (dplyr, tidyr, lubridate, ggplot2), AFT Fathom, ASPEN+, AutoCAD
- Fundamentals and Principles of Engineering (FE, PE) exams passed